

ECON 351 · Focused Practice Set

Set 3 of 10

Chapter 4

Intertemporal Consumption

2 questions in this practice set

Questions: Q5, Q6

Screenshots from original HW, sample questions, and mock midterm.

Question 1 of 2 (Practice Q5)

Intertemporal #1 · Sample Q41

Question 38:

A consulting company researched the market for apples in California, and concluded the following: apples and bananas are complements, and apples are an inferior good.

Define the following:

Q_A : quantity of apples demanded

P_A : price of Apples

P_B : price of bananas

I : income

From the options below, mark the only demand function that can possibly represent the demand for apples in California.

(A) $Q_A = 100 - 3P_A + 3I - 5P_B$

(B) $Q_A = 200 - P_A + 2I + 4P_B$

(C) $Q_A = 110 - 2P_A - 4I + 2P_B$

(D) $Q_A = 250 - 4P_A - 5I - 3P_B$

(E) $Q_A = 150 + P_A + I + P_B$

Question 39:

The demand for apples (Q_A) is given by the function

$$Q_A = 784 - 2P_A - 3I - 4P_O$$

where P_A is the price of apples, I is the income level, and P_O is the price of oranges.

Suppose $P_A = 2$, $I = 100$, and $P_O = 20$.

a) Compute the price elasticity of demand for apples.

b) Compute the income elasticity of demand for apples. Are apples a normal good or inferior good?

c) Compute the cross-price elasticity of demand for apples. Are apples and oranges substitutes or complements?

Question 40:

The income elasticity of demand for good X is -2. Due to a financial crisis, the income of consumers declined by 2%. How did the financial crisis affect the quantity demanded for X?

(A) The quantity demanded increased 2%

(B) The quantity demanded decreased 2%

(C) The quantity demanded increased 4%

(D) The quantity demanded decreased 4%

(E) The quantity demanded did not change

Question 41:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 50%, that is, $r = 0.5$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

What is the optimal choice of q_1 and q_2 ? Is this consumer saving or borrowing? How much?

Question 42:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 5%, that is, $r = 0.05$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

What is the optimal choice of q_1 and q_2 ?
Is this consumer saving or borrowing? How much?

Question 2 of 2 (Practice Q6)

Intertemporal #2 · Mock Q24-25

Question 24:**[Show your work]**

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 60$ in the first period and $I_2 = 125$ in the second period. The consumer can borrow or save money at the interest rate of 25%, that is, $r=0.25$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

The consumer's inter-temporal budget constraint is

- (A) $q_1 + 0.8q_2 = 160$
- (B) $q_1 + q_2 = 185$
- (C) $q_1 + 1.25q_2 = 216.25$
- (D) $q_1 + 0.75q_2 = 160$
- (E) $q_1 + 0.25q_2 = 200$

Question 25:**[Show your work]**

Consider all the information from Question 24. In the first period, the consumer will optimally purchase

- (A) $q_1 = 0$
- (B) $q_1 = 100$
- (C) $q_1 = 60$
- (D) $q_1 = 80$
- (E) $q_1 = 120$